

Project Proposal: Smart IoT-Based Weather Monitoring System

Group Number : 02

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1 Introduction

Current environmental monitoring solutions often struggle with the balance between high fidelity data collection and energy autonomy. This project proposes an industrial grade, self sustaining weather station built on the ESP32 platform. The primary objective is to investigate and implement advanced **Low Power Optimization techniques** that allow the system to operate indefinitely. (using harvested solar energy.)

2 Core Objectives

- **Energy Autonomy:** Implementation of ultra low power deep sleep cycles ($< 10\mu A$) and hardware-level power and optimized charging circuits..
- **Hybrid Connectivity:** Development of a communication stack that adapts between Wi-Fi and GSM/GPRS based on signal availability and power constraints.
- **Data Integrity:** Efficient local buffering in RTC memory to minimize power-intensive radio transmissions, combined with idempotent API ingestion to prevent data duplication..
- **Scalable Backend:** A high performance time-series data ingestion API using FastAPI and TimescaleDB for efficient time series data storage and analysis..

3 System Architecture

3.1 Hardware Layer

The hardware is designed as a modular bus system, allowing for interchangeable sensors based on specific deployment needs.

- **Processing Core:** ESP32 (utilizing dual-core architecture for separation of sensing and transmission tasks).
- **Environmental Sensing:** Support for I2C based sensors such as the BME280 (Temperature, Humidity, Pressure) and optical sensors like BH1750.
- **Power Subsystem:** A modular solar to lithium charging circuit using a triple solar panel array (approx. 3W peak) and 18650 cells. Power is distributed via a Low Dropout Regulator (LDO) to maintain efficiency.

3.2 Software & Power Strategy

Rather than constant polling, the system employs a "**Sample, Buffer, Burst**" strategy:

1. **Sample:** Sensors are powered on via a MOSFET switch, a reading is taken, and power is immediately cut.

2. **Buffer:** Readings are stored in the ESP32's RTC memory, which persists during deep sleep.
3. **Burst:** Once the buffer reaches a threshold, the system initiates a single high speed transmission (Wi-Fi or GPRS) to the backend.

4 Industrial-Grade Backend Design

To ensure the system is production ready, the backend follows modern microservice standards:

- **API Framework:** FastAPI or Express/Fastify for asynchronous, high concurrency data ingestion.
- **Database:** Using a database that is optimized for time series environmental data like TimescaleDB (PostgreSQL)
- **Security:** Device-level authentication using unique API keys and rate-limiting to prevent ingestion floods. Add Schema Validation and Idempotent Ingestion.
- **Observability:** Comprehensive health checks and threshold based alerting systems.

5 Methodology & Build Order

The project will follow an iterative development cycle to ensure stability at each layer:

1. **Phase I: Core Logic.** Establish stable I2C sensor communication and implement deep sleep logic.
2. **Phase II: Power Gating.** Integrate P-channel MOSFETs to completely de-energize sensors during sleep cycles.
3. **Phase III: Networking Failover.** Develop the software logic to switch between WiFi (primary) and GSM (backup) protocols.
4. **Phase IV: Data Visualization.** Deploy the FastAPI backend and connect a dashboard (e.g., Grafana) for real time monitoring.

6 Feasibility & Sustainability

Given the use of an ESP32, the system is well equipped to handle complex networking protocols. By utilizing the solar rechargeable battery system, the energy budget allows for sampling time intervals (about 15 -30 minutes) while maintaining a positive battery charge even during low light conditions.

7 Conclusion

This project serves as a research prototype for highly efficient, remote environmental monitoring. By focusing on modularity and power optimization rather than fixed components, the proposed framework can be adapted for agriculture, disaster management, or urban smart city initiatives.